June 17, 2026

To the Editorial Board
Journal of Statistical Mechanics: Theory and Experiment (JSTAT)

Dear Editors,

Please consider the enclosed manuscript, "A two-feedback Vicsek–Couzin flock: dense-phase noise rectification as the unifying action of density-gated motility–noise coupling," for publication in JSTAT as an original research article.

The standard Vicsek model fixes the self-propulsion speed and the angular-noise distribution as global constants. Two relaxations have been studied separately: a density-dependent speed, which drives motility-induced phase separation, and heavy-tailed Lévy reorientations, which reshape the order–disorder transition. We ask what happens when both are switched on together, gated by a single density-dependent sigmoid in the local neighbour count, and we isolate each contribution with a four-way ablation (baseline / noise-shape-only / motility-only / full) run on one code path.

We find that heavy-tailed noise on its own grows no size-dependent susceptibility; only the motility channel builds a phase-separated dense phase. The noise-shape channel then acts inside that dense phase through a single route, rectifying its heavy-tailed Cauchy kicks into Gaussian ones. What at first looks like a super-additive cooperation between the two channels we show, with a matched-noise control, to be one effect: equalising the effective dense decorrelation collapses the susceptibility excess (Δ_9 from +0.45 to +0.02) and reverses the static $g(r)$ and dynamic $C(\tau)$ gaps, so the two channels are additive and the noise-shape feedback simply lowers the dense effective noise. The robust results are a clean two-regime split that reproduces across three densities, a density-gated Lévy-to-Gaussian conversion of dense-phase turning, and a decoupled-sigmoid dissection that pins the motility threshold as the sole control variable and shows the effect to be a property of the metric kernel.

We believe the work fits JSTAT squarely. It is a statistical-mechanics study of an active-matter model — Vicsek and Couzin flocking, motility-induced phase separation, giant number fluctuations, finite-size scaling of the order–disorder transition, and density-stratified correlation functions — and should interest the active-matter and non-equilibrium statistical-mechanics communities the journal serves.

The analysis is deliberately conservative. Claims rest on ten-seed finite-size scans with seed-bootstrap confidence intervals and non-parametric (Wilcoxon signed-rank, Mann–Whitney) tests rather than Gaussian z-scores; null results (a passive noise-shape threshold, negligible hysteresis, no extended patch lifetime, a topological-kernel breakdown, and a falsified mean-density prediction) are reported rather than buried; the finite-size slopes are presented as crossover rates, not asymptotic exponents, dropping as the lever is extended to nine sizes (L up to 256); and the apparent synergy is shown, by the matched-noise control, to be additive at matched noise rather than an emergent cooperation. All code and data are openly available at https://github.com/yayayou47/two-feedback-vicsek for bit-for-bit reproduction of every figure.

This manuscript is original, has not been published previously, and is not under consideration at any other journal. It has a single author, who declares no competing interests. One supplementary-material PDF accompanies the submission. As potential referees, we would suggest specialists in active-matter and flocking models, in finite-size scaling of non-equilibrium phase transitions, and in motility-induced phase separation.

Thank you for considering our submission.

Sincerely,

Yaya Youssouf Yaya
Département de Mathématiques, École Normale Supérieure de N'Djaména, Tchad
Chercheur Associé, LMA, Université Assane Seck de Ziguinchor, Sénégal
yy.y@zig.univ.sn ORCID: 0000-0003-0781-4923